

AKSHAYA AYYAH

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EDUCATION

University of Illinois Urbana Champaign(Stamps Scholar)

Expected Graduation: May 2028

Bachelors of Science in Computer Science + Chemistry; Minor in Bioengineering, Dance

GPA: 3.96/4.00

- *Awards:* Selected for the nationally competitive, four year, full-ride "Stamps Scholarship" at UIUC, chosen on the merits of leadership, academics, and service
- *Coursework:* Bioinformatics, Computational Biology, Biochemistry, Data Science for Chemical Engineering, Data Structures, AI/ML, Linear Algebra, Molecular Genetics, Algorithms, Biostatistics, Software Design
- *Involvement:*
 - MedLaunch, *Outreach Director:* Applied engineering principles to create assistive devices for individuals with disabilities, networked with community partners to find suitable projects
 - Biomedical Engineering Society, *Marketing Chair:* Designed and curated content for marketing purposes and recruitment

EXPERIENCE

Proctor & Gamble, Research and Development Intern

May 2026 - Aug 2026

- Built a neural-network-powered regulatory compliance system leveraging web crawlers, REST APIs, NLP pipelines, and chemical informatics data to identify how substance-level compliance status, potential formulation compliance gaps, and regulatory forecasts impact product portfolio
- Partnered with regulatory managers, toxicologists, environmental scientists, and R&D teams to translate complex chemical safety and regulatory requirements into scalable technical specifications, predictive models, and decision-support tools used across product development workflows

Argonne National Laboratory, Research Aide

May 2025 - Present

- Used neural networks to generate and improve the design of antibodies/minibinders targeting potential cancer target proteins as a full time intern with the Ramanathan Lab
- Utilized machine learning models/techniques such as diffusion modeling, preference optimization, large language models, molecular dynamics simulations, and reinforcement learning to model novel antibody designs

University of Illinois Urbana Champaign, Undergraduate Researcher

January 2025 - Present

- Utilized rapid chemical imaging and computational imaging of bladder and endometrial tumor tissue, in order for accurate and efficient classification of malignant tissue as part of the Bhargava Lab at the Cancer Center of Beckman Institute

PROJECTS

VisionSwim

October 2025 - May 2026

- Device using computer vision to alert a swimmer with vision loss when he swims off-course so that he can re-angle his direction

IlliniSearch

August 2025 - December 2025

- AI-powered search engine designed specifically for the UIUC community which aggregates and indexes content from key UIUC-related platforms - including Reddit, Rate My Professor, Course Explorer, grade disparity tools, general education databases, and official university websites - to provide students with fast, centralized access to campus-related information.

ProjectPink, Awarded Best Web Development at CodeAda 2025

October 2025

- A scrollytelling web experience that exposes the Pink Tax, with camera detection functionality to detect "pink taxed" product and offer alternatives
- Built using React, JS, and CSS-powered animation inspired by 80s aesthetics and Apple-style storytelling.

SpinaCare

October 2024 - May 2025

- Collaboratively engineered an adaptive and accessible toilet seat for a eight-year-old child with *spina bifida*
- Applied engineering principles to brainstorm, design, build, test, and improve the device based off of the community partner's feedback
- Successfully delivered device to the community partner

SKILLS

Languages: English, Tamil, Spanish

Coding Languages: C++, C, Java, Java Script, HTML, Python, React